

LECTURE 2

LAST TIME

LECTURE 1: GREAT TRADES

Paulson's bet against subprime (CDS)

Soros's bet against the pound (currency futures)

IBM World Bank swap (swaps)

Thales' bet on oil (options)

THIS TIME

LECTURE 2: OUTLINE

Empirical evidence on derivatives use

Arbitrage?

HISTORY

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70s: FX futures, equity options, T-bill futures

80s: FX swaps, IR swaps, options on futures, swaps, Asian options

90s: Exotics, catastrophe options, MBS

00s: Volatility, credit/energy/weather derivatives, CDOs

BIG MARKET GREW RAPIDLY

Notional value of outstanding derivatives c. 620 trilli USD (2023)

C. 250 times more than in 1980

DERIVATIVES DATA FROM BIS

Growth of total notional value

Notional (nominal or face amount) $\neq$ mkt value

Net mkt value = 0 (short for every long)

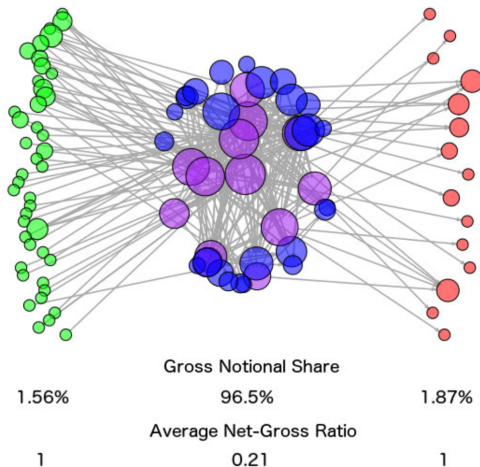
Can be grouped by underlying

Equities, commodities, interest rates, credit,...

Can be grouped by type of instrument

Futures, options, swaps,...

NET VS. GROSS IN CDS



See D'Errico–Roukny (2017)

BUT TO USE DERIVATIVES, YOU HAVE TO PRICE THEM

HOW DO YOU PRICE DERIVATIVES?

Price derivatives by the principle of no arbitrage

An arbitrage is a portfolio that either

- (i) Costs nothing and today and yields profit in future
- (ii) Yields profit today and costs nothing in future

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No arbitrage exists in “well functioning markets”

Else, arbitrageurs can make infinite profit

No arb like a physical law of derivatives pricing

QUICK PRACTICE

Suppose Bond A and Bond B each pay off \$1 tomorrow

(Say “bond” for zero-coupon bond w/ \$1 face value)

Bond A costs 95¢ and Bond B costs 97¢

Is there arbitrage?

What is it?

GENERAL ARB WITHOUT UNCERTAINTY

Let V_t be the value of a portfolio at date t

V_0 denotes its value today

V_1 denotes its value tomorrow

Suppose no uncertainty: V_1 known at date 0 (as in bond example)

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When is the portfolio with value V an arbitrage?

PORTFOLIO WITH NO UNCERTAINTY

today

tomorrow

V_0



V_1

ARBITRAGE WITH NO UNCERTAINTY

With no uncertainty, V is an arbitrage if either

- (i) $V_0 < 0$ and $V_1 \geq 0$ or
- (ii) $V_0 = 0$ and $V_1 > 0$

ARBITRAGE WITH UNCERTAINTY

Now suppose simplest form of uncertainty

Suppose two “states of nature” (or “states”) tomorrow

state $\omega = u$ (for “up”)

state $\omega = d$ (for “down”)

Let V_ω be the value of the portfolio V tomorrow in state ω

ARBITRAGE WITH UNCERTAINTY

Now suppose simplest form of uncertainty

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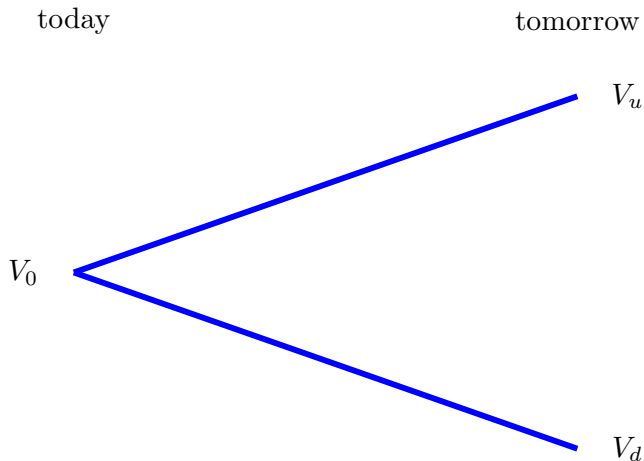
state $\omega = u$ (for “up”)

state $\omega = d$ (for “down”)

Let V_ω be the value of the portfolio V tomorrow in state ω

When is the portfolio with value V an arbitrage?

A PORTFOLIO WITH UNCERTAINTY



ARBITRAGE WITH UNCERTAINTY

With uncertainty over $\omega \in \{d, u\}$, V is an arbitrage if either

(i) $V_0 < 0$ and both $V_u \geq 0$ and $V_d \geq 0$ or

(ii) $V_0 = 0$ and either

$V_u > 0$ and $V_d \geq 0$ or

$V_u \geq 0$ and $V_d > 0$

PRINCIPAL OF THE ABSENCE OF ARBITRAGE

NO ARBITRAGE (NA)

What do you do if V is an arbitrage portfolio?

If an arbitrage exists, demand for it infinite

Supply cannot equal demand—“markets cannot clear”

Well-functioning markets $\implies$ no arbitrage

LAW OF ONE PRICE

LAW OF ONE PRICE (LoOP)

Portfolios with equal payoffs have equal prices

In notation: let V and W be values of two portfolios

$$V_1 \equiv W_1 \text{ (equal payoffs tomorrow (for each } \omega))$$

$$\implies V_0 = W_0 \text{ (equal prices today)}$$

NA $\implies$ LoOP

Proof by contradiction: suppose NA, $V_1 \equiv W_1$, and $V_0 > W_0$

Form portfolio: long W and short V

Payoff tomorrow: $W_1 - V_1 \equiv 0$ (by assumption)

Cash flow today: $-W_0 + V_0 > 0$

Arbitrage! Contradicts hypothesis $\implies V_0 \not\geq W_0$ ($V_0 \leq W_0$)

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HW: Show likewise $V_0 \geq W_0$ to conclude $V_0 = W_0$

LoOP IS BACKBONE OF COURSE

LoOP gives a cookbook for pricing many many derivatives

Replicate derivative payoff with securities portfolio

Derivative has same price as portfolio

This course: learn and practice recipes in cookbook

LoOP EXAMPLES

Two stocks, S^1 and S^2 , have prices $S_0^1 = 1$ and $S_0^2 = 2$

1. What is price of portfolio with one unit of S^1 and one of S^2 ?
2. What is price of derivative that pays off sum of prices $S_1^1 + S_1^2$?
3. Is derivative that pays $\max\{S_1^1, S_1^2\}$ worth more or less than 1? Than 2?

BUT...

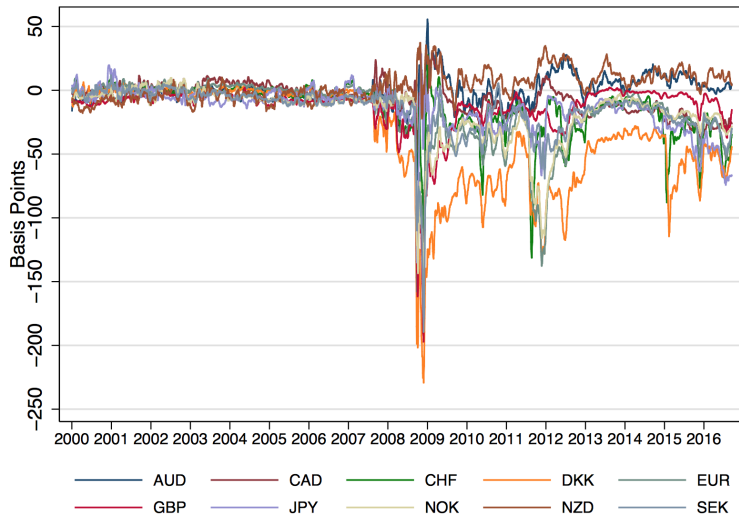
Markets not “well-functioning” all the time

Sometimes arb could be possible

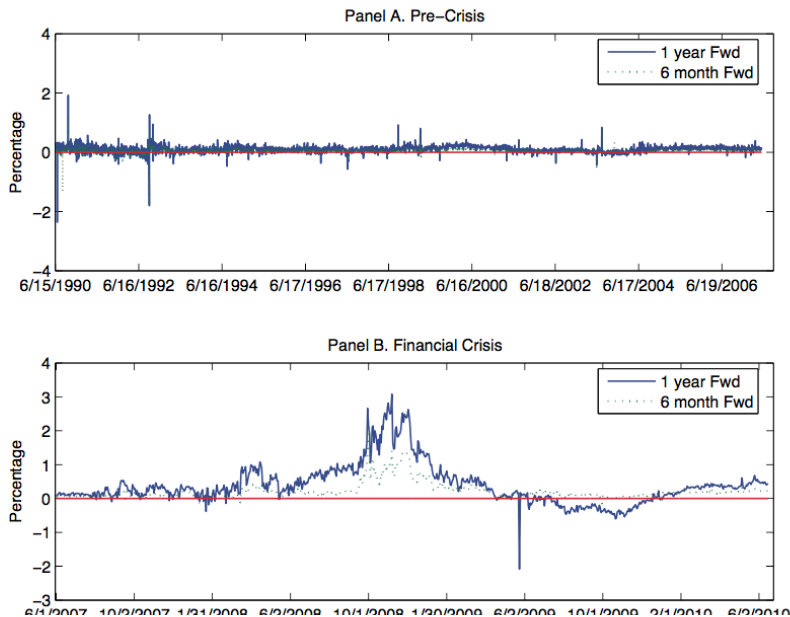
E.g. post crisis arb in exchange rates

Violations of covered interest parity (CIP) (more soon)

CIP VIOLATIONS



CIP VIOLATIONS



STILL, HOW MUCH IS CIP VIOLATED BY?

“Big” violations to arbitrageurs smallish to normal traders

Can't avoid credit/counterparty risk and transactions costs

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Can't avoid credit/counterparty risk and transactions costs

But why don't banks eat up those profits?

WHY MIGHT NO-ARB FAIL?

Lack of borrowing capacity/debt capacity/holding capacity

Need to short to arb

Need to borrow to arb

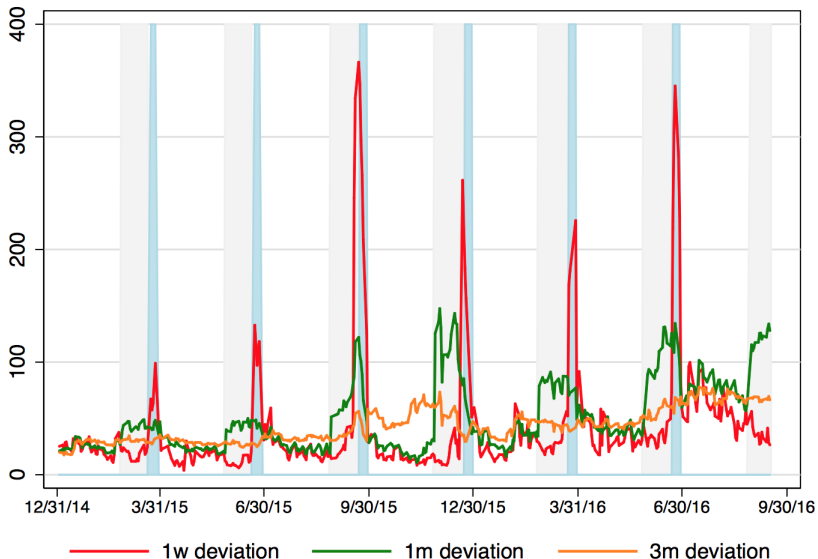
Need to hold positions (clients might withdraw before arb)

NO-ARB FAILING DUE TO REGULATION

“We find that deviations from the covered interest rate parity condition (CIP) imply large, persistent, and systematic arbitrage opportunities in one of the largest asset markets in the world. Contrary to the common view, these deviations for major currencies are not explained away by credit risk or transaction costs. They are particularly strong for forward contracts that appear on the banks’ balance sheets at the end of the quarter, pointing to a causal effect of banking regulation on asset prices”

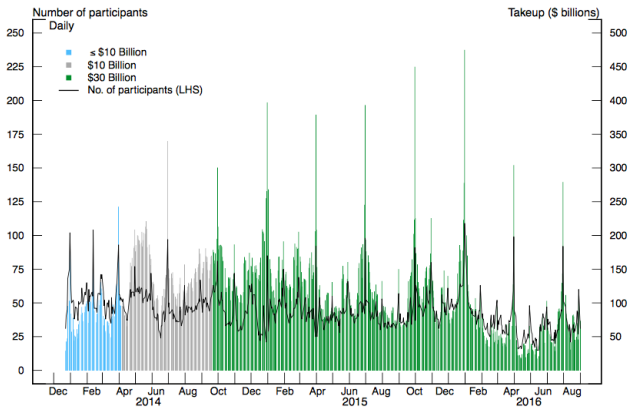
—Du–Tepper–Verdelhan (2017)

REGULATION: REPORTING QUARTER END



WHY? BANKS EXIT REPO FUNDING POSITIONS

Figure 5: RRP Facility Usage



Source: Anbil-Senyuz 2017

RECAP

LECTURE 2: RECAP

Derivatives facts and data

Huge mkts with many instruments and underlyings

No arb and LoOP—laws of derivatives pricing

Perfect most of the time

Imperfect some of the time

Okay all of the time (so far)

FIN